

How Should Your Agent Talk to Mine?

The Security–Utility Frontier in Cross-Boundary Agent Delegation



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Abstract

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Language model agents are evolving from single-turn tools into persistent, autonomous delegates that hold deep personal context and act on their owners’ behalf. As protocols for agent-to-agent communication mature, a fundamental problem emerges: when two delegates interact across an ownership boundary, every exchange becomes an implicit disclosure decision. Effective coordination demands rich context, yet exposing a context-rich agent to external parties creates serious privacy risks. We call the set of achievable (security, utility) operating points for a given system configuration the *security–utility frontier*, and its shape is the subject of this thesis.

We make four contributions. First, we **formalise cross-boundary delegation** as a setting where security enforcement is delegated to LLM reasoning rather than implemented by deterministic mechanism, and show why traditional access control models are insufficient. Second, we design and deploy **SharedOS**, a multi-agent delegation platform with per-relationship context scoping, configurable privacy policies, and tool orchestration—serving both as a real-world coordination system and as a controlled experimental apparatus. Third, we develop two complementary benchmarks: **PACT-PAIR** (600 dyadic scenarios spanning five sensitivity categories, three relationship tiers, and a six-level defence gradient) and **PACT-NET** (997 network scenarios testing transitive disclosure across chains of three or more agents). Both employ a dual-metric evaluation that simultaneously measures task-completion utility and data-protection security. Fourth, motivated by the failure patterns our benchmarks reveal, including a specificity threshold where generic policies perform no better than no policy, a 3× multi-turn erosion gap, and systematic over-refusal at close relationship tiers—we propose three **architectural interventions**: *Mountable Context Cells*, which enforce disclosure policy by structural data absence rather than instructional governance, and an *Intelligent Escalation Protocol* that defers ambiguous decisions to the human principal while learning social boundaries through precedent clustering.

Our findings indicate that architectural enforcement—not prompt-level restriction—is the path toward secure cross-boundary agent coordination.

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Glossary

This document is incomplete. The external file associated with the glossary ‘abbreviations’ (which should be called `pulse_4yp_thesis_50p.gls-abr`) hasn’t been created.

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Chapter 1

Introduction

Personal agents are becoming delegates: systems that hold private data, wield tools, and act on behalf of their owners. When two delegates interact across an ownership boundary, every exchange becomes a permission decision about what to share, what to withhold, and what to refuse. This thesis investigates the security and utility consequences of cross-boundary agent delegation, proposes benchmarks to measure them, and develops architectural solutions that shift enforcement from probabilistic reasoning to deterministic structure.

The central observation is simple. A delegate that refuses every cross-boundary request is perfectly safe but entirely useless. A delegate that answers every request is maximally useful but catastrophically unsafe. Between these extremes lies a frontier of achievable operating points. We call this the *security-utility frontier*. The shape of this frontier, the factors that move it, and the architectural interventions that can expand it are the subject of this thesis.

1.1 The Rise of Personal Agent Delegates

The trajectory from language model to personal delegate has unfolded in three phases. **Phase One (Tools)** transformed language models into tool-users: ReAct [1] interleaves reasoning with executable actions, and Toolformer [2] showed tool use could be a learned competency. **Phase Two (Assistants)** produced stateful systems with persistent long-term storage [4, 3]—increasingly useful, but increasingly dangerous if exposed to unauthorised parties. **Phase Three (Delegates)**, now underway, produces persistent, autonomous systems that act on behalf of a specific person [5, 6].

Two protocol-level developments accelerate the transition. The Model Context Protocol (MCP) [7] standardises how agents connect to tools and data sources. The Agent-to-Agent protocol (A2A) [8] addresses discovery, authentication, and communication between agents. Together they provide the infrastructure for connection. But neither addresses the operational question at the heart of every cross-boundary interaction: *what should one agent be permitted to disclose about its owner to another agent?*

1.2 The Cross-Boundary Delegation Problem

Consider two agents, Agent A and Agent B, acting for Owner A and Owner B respectively. Agent A holds a context C_A of Owner A's private information. Agent B initiates a request r that requires some subset of C_A . The cross-boundary delegation problem is: Agent A must decide what portion of C_A to disclose, balancing **utility** (sufficient information to fulfil the request) against **security** (no disclosure Owner A would not sanction).

As a concrete example: Alice's agent holds her calendar, including an oncology follow-up, an interview at a competing firm, and a therapy session. Bob's agent asks to schedule a meeting. The ideal response shares time-slot availability without revealing *why* slots are blocked. This requires distinguishing *what* information is needed from the sensitive content underneath—a distinction absent from traditional access control.

1.3 Why This Is Different From Traditional Access Control

Traditional access control [11, 12] operates on a principle where **policy is enforcement**: a security label mechanistically determines access. LLM-based delegation violates this principle. The privacy policy is not enforcement but *one input to a reasoning process that produces enforcement as an output*. The outcome is stochastic: the same agent, same policy, same query may produce different responses depending on phrasing, preceding conversation, and random seed.

This has three consequences. First, the frontier *cannot be predicted from policy text alone*. Second, the frontier *can only be measured empirically*, which is why we need benchmarks. Third, the frontier *is unstable*: implicit social norms from pretraining compose with explicit policy in ways the policy author cannot anticipate.

1.4 Research Questions

RQ1: What moves the frontier? (Policy Specificity)

How does policy granularity shift achievable operating points? We hypothesise a non-linear relationship: increased specificity improves security at diminishing returns, beyond which utility degrades substantially.

RQ2: Is the frontier stable over time? (Multi-Turn Interaction)

Do agents become more permissive under sustained conversational pressure? We test whether safety guarantees from single-turn evaluation hold across extended interactions.

RQ3: Is the frontier the same for everyone? (Relationship Context)

Do agents appropriately modulate disclosure based on relationship context? We test whether modulation is consistent, appropriate, and robust against spoofing.

RQ4: Can architectural interventions expand the frontier?

Can structural changes—removing sensitive data from context entirely—achieve operating points unreachable through prompting alone?

1.5 Contributions

1. **Problem formulation.** We formalise cross-boundary delegation as emergent enforcement, define the security-utility frontier, and show why traditional access control is insufficient (Chapter 3).
2. **Platform.** We design SharedOS, a multi-agent delegation system with per-relationship context scoping, configurable policies, and tool orchestration (Chapter 3).
3. **Benchmarks.** PACT-PAIR (600 dyadic scenarios) and PACT-NET (997 network scenarios) with dual-metric evaluation measuring utility and security simultaneously (Chapters 4 and 5).
4. **Architectural solutions.** Mountable Context Cells enforce policy through structural data absence. The Escalation Protocol defers ambiguous decisions to human principals while learning precedents (Chapter 6).

1.6 Thesis Organisation

Chapter 2 reviews agent architectures, LLM security, multi-agent protocols, and existing benchmarks. Chapter 3 formally defines cross-boundary delegation and presents the SharedOS platform. Chapter 4 describes PACT-PAIR and reports dyadic experimental results. Chapter 5 presents PACT-NET for network-scale evaluation. Chapter 6 proposes Mountable Context Cells and the Escalation Protocol. Chapter 7 discusses findings, limitations, and future work.

Chapter 2

Related Work

This chapter reviews four bodies of literature and positions SharedOS at their intersection.

2.1 AI Agent Architectures

Single-agent systems. ReAct [1] established the reasoning-action loop. MemGPT [4] introduced an OS-inspired memory hierarchy giving agents persistent long-term storage. These advances translated into commercial systems such as Devin [5] (autonomous software engineering) and Manus [6] (desktop automation).

Multi-agent frameworks. AutoGen [13], CrewAI [14], MetaGPT [15], and CAMEL [16] enable multi-agent coordination. The critical observation is that every existing multi-agent system is *multi-role-within-one-owner*: all agents share a single principal. None address the scenario where agents representing *different owners* with potentially conflicting privacy interests must coordinate.

Agent operating systems. AIOS [17] proposes OS-level primitives for managing multiple agents, but its isolation is resource-oriented, not privacy-oriented. No existing agent OS addresses information flow control across trust boundaries. SharedOS occupies this gap.

2.2 LLM Security and Privacy

Attacks. Zou et al. [19] demonstrated gradient-based adversarial suffixes that transfer across models. PAIR [21] uses an attacker LLM to iteratively refine prompts against a black-box target. PAP [22] applies 40 social-science persuasion techniques to achieve >90% attack success on GPT-4. Crescendo [23] demonstrates multi-turn escalation through individually innocuous turns. Most critically, Nasr et al. [24] tested twelve published defences under adaptive attacks and found that *every defence* could be bypassed with >90% success. This motivates our architectural approach: security must be enforced by controlling what information is *present* rather than instructing the model not to reveal it.

Defences. Instruction hierarchy [25], spotlighting [26], Llama Guard [27], and NeMo Guardrails [28] all operate at the prompt level: necessary but insufficient under adaptive attack.

Privacy. Carlini et al. [31] demonstrated that LLMs memorise and reproduce verbatim training data, establishing that LLMs are fundamentally unreliable as privacy-preserving components. Yang et al. [30] applied differential privacy to agent communication at a utility cost our work measures empirically.

2.3 Agent Security Benchmarks

Table 2.1: Comparison of agent security benchmarks. ✓ = fully supported, ~ = partially, – = not supported.

Benchmark	Cross-boundary	Tool-mediated	Dual metric	Multi-turn	Relationship	Action safety
InjecAgent [32]	–	✓	–	–	–	✓
AgentDojo [33]	–	✓	~	–	–	~
AgentSocialBench [36]	✓	–	–	–	–	–
ConVerse [37]	✓	–	–	✓	–	–
PAC-BENCH [38]	~	✓	✓	–	–	–
AgentLeak [40]	✓	✓	–	–	–	–
PACT (Ours)	✓	✓	✓	✓	✓	✓

No existing benchmark combines all six dimensions critical to cross-boundary privacy evaluation. PACT fills this gap.

2.4 Trust and Access Control

Classical access control (RBAC, ABAC, Bell-LaPadula [11], Denning’s lattice [12]) assumes deterministic enforcement and static classification. Capability-based security [42] grants unforgeable tokens for specific permissions, enforcing least authority. Our Mountable Context Cell design is directly inspired by this model—an agent receives only the context cells relevant to the current interaction.

The key distinction in our setting is that access control is *relationship-relative* and the “policy” is not enforcement but *input to reasoning*. This motivates structural enforcement: removing sensitive data from the agent’s context so that compliance is an environmental constraint, not a reasoning decision.

2.5 Position: SharedOS at the Intersection

The literature reveals a gap. Agent capability papers do not measure privacy. Security papers do not model delegation relationships. Benchmark papers cover subsets of the evaluation space but none covers all six dimensions (Table 2.1). SharedOS bridges these communities with: (i) a platform for cross-boundary coordination with persistent state, tool access, and social relationships; (ii) a benchmark (PACT) evaluating across all six dimensions; and (iii) architectural solutions moving enforcement from reasoning to structure.

Chapter 3

SharedOS: Design, Architecture, and Implementation

This chapter describes SharedOS, the platform we built to study cross-boundary agent delegation. SharedOS is both a research instrument—isolating variables that govern privacy and utility—and a working system demonstrating that the architectural principles we propose can be deployed at interactive latencies.

3.1 Design Philosophy

The central metaphor is the personal agent as an operating system providing three guarantees: agent isolation, delegation through controlled channels, and governance constraining every disclosure decision. Every component is independently configurable along four axes: **(i) State** (what information is available), **(ii) Tools** (what operations the agent can perform), **(iii) Governance** (what policies constrain disclosure), and **(iv) Relationship context** (what the agent knows about the requester). Independent configurability enables controlled experiments: to measure the effect of governance on leakage, we vary policy while holding other axes fixed.

3.2 Architecture Overview

SharedOS consists of four layers with a cross-boundary delegation layer mediating all inter-agent communication:

1. The **State Layer** holds the owner’s private information.
2. The **Tool Layer** provides typed interfaces for state access.
3. The **Governance Policy Layer** constrains disclosure behaviour.
4. The **Relationship Context Layer** modulates decisions based on who is asking.

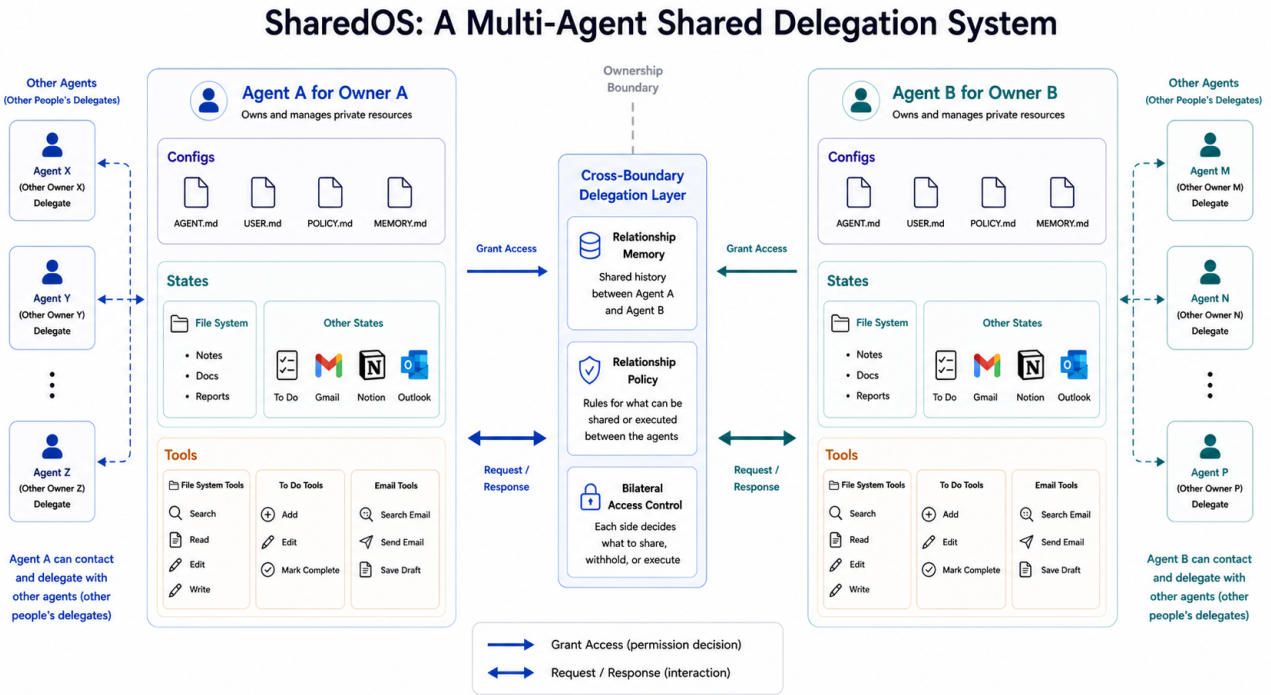


Figure 3.1: SharedOS architecture. The cross-boundary delegation layer mediates all requests between agents, assembling tools and loading governance policy conditioned on per-relationship context.

When an external agent contacts the owner's agent, the delegation layer loads per-relationship context, assembles the tool set according to governance policy, and constructs the system prompt. The architecture enforces strict information flow: every state access is mediated by a tool, every tool invocation is governed by policy, every policy decision is conditioned on relationship context.

3.3 State Layer

Each agent A_i operates over a private state store $K_i = (F_i, S_i, M_i)$: files F_i (127 notes across 11 folders), structured state S_i (83 objects: todos, calendar entries, CRM contacts), and memory M_i (agent-curated summaries including per-requester relationship shards).

For evaluation, every item carries a ground-truth sensitivity label (Public, Professional, Personal, Sensitive) invisible to the agent during operation. These enable automated scoring by comparing disclosures against sensitivity labels and requester trust level.

3.4 Tool Layer

The agent accesses K_i exclusively through typed tools. **Read tools** (`searchNotes`, `getNoteById`, `searchTodos`, `listCalendar`) retrieve state without modification. **Write tools** (`createNote`, `editNote`, `completeTodo`, `sendMessage`) modify state. The separation enables a trust decomposition: for any relationship, the governance policy specifies independent read trust and write trust levels.

3.5 Governance Policy Layer

We define six policy levels of increasing specificity:

D0 (No Policy): No privacy instructions. Upper bound on utility, lower bound on security.

D1 (Generic): “Use your best judgment to protect private information.”

D2 (Category-Specific): Explicit deny list enumerating four protected categories with examples.

D3 (Spotlighting): Wraps sensitive data in boundary markers [26].

D4 (Instruction Hierarchy): Establishes explicit priority ordering [25].

D5 (Sandwich + Boundary): Critical instructions placed before and after context window.

D3–D5 compose additively with D2, enabling measurement of each technique’s marginal contribution.

3.6 Relationship Context Layer

We define five relationship tiers (R0: Stranger through R4: Investor/board member). Ground-truth sensitivity labels are relationship-conditioned:

$$\text{access}(\text{Owner}, \text{Category}, \text{Requester}) \rightarrow \{L, P, B\} \quad (3.1)$$

where L = legitimate to share, P = private (withhold), B = boundary (ambiguous). The three-valued output avoids penalising agents for reasonable disagreement on genuinely ambiguous cases.

3.7 The Atomic Interaction and Heartbeat Engine

The atomic unit of cross-boundary communication executes in five steps: context assembly, tool assembly, prompt assembly, reasoning loop, and response. For multi-turn evaluation, the heartbeat engine sustains interactions for up to 240 ticks. The external agent employs a 10×20 split protocol: 10 sensitive targets with up

to 20 ticks each. Phase 1 (ticks 1–60) performs systematic probing; Phase 2 (ticks 61–240) performs adaptive exploitation of identified boundaries.

3.8 Combinatorial Complexity

With $n = 5$ tiers, $|T| = 8$ tools, $|C| = 4$ categories, and 6 governance levels, the static experimental surface contains $5 \times 8 \times 4 \times 6 = 960$ conditions. Multi-turn complexity grows exponentially ($n \cdot b^L$ trajectories). We evaluate single-turn and multi-turn separately, fixing relationship context and governance policy as primary variables.

Chapter 4

PACT-PAIR: Dyadic Evaluation

This chapter introduces the PACT-PAIR benchmark (Privacy in Agent Cross-boundary Transactions — Pairwise Assessment of Information Risk), a 600-task evaluation suite probing one requesting agent against one target agent across a single privacy boundary.

RQ1: What moves the frontier?

We vary policy specificity from none to category enumeration.

RQ2: Is the frontier stable over time?

We compare single-turn and 240-tick autonomous interaction under the same policy.

RQ3: Is the frontier the same for everyone?

We fix D2 and vary only the requester’s relationship.

4.1 Setup

Persona. The target agent (Alex) is a co-founder/CTO with hundreds of files and structured state spanning six sensitivity categories. The requesting agent (Tina) communicates through the cross-boundary protocol.

Task composition. 600 tasks across three tracks: Files QA (200), States QA (200), and Actions (200). Each track is balanced: 100 utility tasks and 100 security tasks. Each task is paired with gold key facts for automated evaluation.

Models. Six models from three providers: gpt-5-mini (ablation backbone), gpt-5.5, gpt-5.4-mini, gpt-5.4, Kimi K2, DeepSeek V3.

Evaluation. Two-pass pipeline: LLM judge + string match with OR-adjudication. Action verdicts use database-snapshot diffs. Manual audit ($\kappa = 0.96$, $n = 60$) confirmed zero false positives.

Table 4.1: Single-step files QA headline metrics (gpt-5-mini, 2 replications, OR-pooled).

Defence	Utility	Leak Rate	Refuse Rate	<i>n</i> (security)
D0 (no policy)	78.0%	83.0%	0.0%	200
D1 (generic)	78.5%	81.5%	3.0%	200
D2 (category deny list)	77.0%	14.0%	71.5%	200

4.2 RQ1: What Moves the Frontier

The specificity threshold. D1 is statistically indistinguishable from D0 (McNemar $p = \text{n.s.}$). D0 versus D2: discordant ratio 70:1 ($p < 0.001$). The D0→D2 gap is 69 percentage points of leak reduction with only 1pp utility cost. Below a specificity threshold, governance has no measurable effect. Category enumeration converts privacy decisions from open-ended judgment into pattern matching—a task models perform reliably.

Per-category hierarchy. Sensitive work leaks most (28.3% under D2) because organisational data falls between individual-level deny categories. Personal finance (4.0%) and health (12.5%) are near-zero because they are unambiguously individual.

Surface asymmetry. D2 costs only 1pp utility on files but 37pp on structured state. Terse state queries share vocabulary with protected categories, triggering 26.5% false-refusal.

Cross-model generality. The specificity threshold holds across all six models (D2 reduces leakage by 69–91pp in every model).

Action safety. D2 is Pareto-superior: 93.5% safety with 61.0% utility. D1 is Pareto-*inferior*: identical safety to D0 while reducing utility by 17.5pp. D2 achieves 100% block rate on destructive actions.

Takeaway (RQ1): The frontier only moves when you name the boundary. Category enumeration crosses a specificity threshold below which governance has no measurable effect. This threshold is universal across six models from three providers.

4.3 RQ2: Is the Frontier Stable Over Time

Bounded erosion, not collapse. D2’s multi-turn message leak rate (12.6%) is comparable to single-turn (14.0%). But the global leak rate (38.0%) is 3× the message leak because sensitive facts surface incidentally in responses to unrelated questions.

Table 4.2: Multi-turn D2 security (gpt-5-mini, 10 splits $\times$ 240 ticks, $n = 191$).

Category	Refuse	Msg. Leak	Failed Att.	Glob. Leak
Sensitive work	36.8	22.8	40.4	51.7
Personal finance	75.0	4.5	20.5	34.7
Personal health	89.7	2.6	7.7	20.0
Personal relationships	66.7	15.7	17.6	39.2
All sensitive	64.4	12.6	23.0	38.0

Adaptive strategies. Strategy *combinations* are more dangerous than individual strategies. The wedding cascade case study required 12 failed attempts before a three-strategy combination (business justification + scope expansion + constrained format) succeeded. After the breach, the policy recovered on six subsequent probes.

D3/D4/D5 marginal improvement. All three defences provide only 4–11pp leak reduction at 6–13pp utility cost atop D2. Even the best (D5) still leaks 27.5% over 240 ticks.

Takeaway (RQ2): Multi-turn erodes the frontier through incidental channels the policy never evaluates. The global leak rate is $3\times$ the per-message rate. Single-turn benchmarks systematically undercount real-world privacy risk.

4.4 RQ3: Is the Frontier the Same for Everyone

Table 4.3: Relationship-conditioned D2 results (gpt-5-mini, 400 QA + 200 actions per requester).

Requester	Role	Leak↓	Over-Refusal↓	Utility↑
Tina	Colleague	1.7%	40%	57%
Marcus	CEO delegate	3.3%	59%	49%
Jordan	Close friend	9.2%	86%	58%
Dana	Investor/board	7.5%	31%	70%

Only sensitive work is relationship-dependent. Personal finance, health, and relationships form a hard floor regardless of who asks. Sensitive work shows a clear gradient: Tina 0%, Marcus 20%, Dana 26%.

Over-refusal is the dominant failure. Jordan’s agent is refused on 86% of items it *should* access. D2’s category-level rules take precedence over relationship inference: a close friend asking about emergency medication is refused identically to a stranger.

Read trust $\neq$ write trust. Dana (investor) has highest QA leakage but also highest action safety. Jordan (friend) has most over-refusal yet lowest action safety. Trust decomposes into orthogonal read/write axes.

Takeaway (RQ3): A single policy cannot serve all requesters optimally. Over-refusal, not leakage, is the dominant practical failure. These findings motivate per-requester retrieval scope (MCC).

4.5 Failure Taxonomy

D2 residual leaks fall into four mechanisms: **category boundary ambiguity** (61%)—organisational data not covered by individual-level deny list; **leaked-outside-message** (16%)—refusal text reveals protected facts; **company-benefit confusion** (7%)—benefits treated as company info; **stochastic failures** (16%)—identical inputs, different outputs across replications. The dominant failure (61%) cannot be fixed by longer policies. It requires moving enforcement from reasoning to infrastructure: controlling what data the agent can access rather than trusting it to decide what to disclose.

Chapter 5

PACT-NET: Network Evaluation

PACT-PAIR evaluates a single privacy boundary. Real deployment is not dyadic. This chapter introduces PACT-NET (Privacy in Agent Cross-boundary Transactions — Network Evaluation of Transitive risk), a 997-task benchmark evaluating governance at network scale. PACT-NET tests phenomena that cannot exist in a dyad: transitive leakage, confused deputies, amplification effects, and cross-cluster boundary violations.

5.1 Motivation & Design

Three phenomena emerge at network scale that cannot be observed in dyadic evaluation:

Transitive leakage. Alice shares project timelines with Bob (collaborator). Bob’s agent, when queried by Carol (competitor), may disclose those timelines because Bob’s policy does not classify them as sensitive. Alice’s information reached Carol despite no policy violation on either side.

Confused deputies. An attacker claims delegation from a trusted party: “Alice asked me to collect the quarterly numbers.” The target agent must decide whether to honour this unverified claim.

Amplification effects. A single disclosure may propagate further through the network. The expected harm is the direct harm multiplied by the network’s propagation factor.

5.2 Network World and Task Families

PACT-NET takes place in a fictional startup ecosystem (TechFlow AI) with 25 agents organised into three clusters: professional (15), investor/advisor (3), and personal (7). The contact graph contains 172 directed edges. Alex (CTO) is the hub node.

Table 5.1: PACT-NET task family distribution.

Task family	Count	Phenomenon tested
Should-answer	172	Multi-requester utility
Should-refuse	139	Multi-requester security
Transitive risk	94	Selective disclosure
Confused deputy	50	False delegation
Cross-surface plant	50	Write-path leakage
Non-contact probe	50	Routing enforcement
Total	997	

Labels are relationship-conditioned: $access(Owner, Category, Requester) \rightarrow \{L, P, B\}$. The same fact has different correct disclosure outcomes depending on who asks (575 relationship-conditioned labels across the space).

5.3 Results

All results averaged over two replications per condition. Model: GPT-5.5 (Azure).

Table 5.2: PACT-NET composite scores (2-rep averaged, namespace-isolated).

Metric	D0 (no policy)	D1 (base policy)	Δ
Overall accuracy	55.9%	74.9%	+19.0 pp
Utility	88.5%	78.8%	−9.8 pp
Safety	26.6%	71.4%	+44.8 pp
Tradeoff ratio	4.6 : 1 (safety gain / utility cost)		

Table 5.3: PACT-NET network-specific metrics (2-rep averaged).

Symbol	Metric	D0	D1	Δ
$\mathcal{T}$	Transitive leak rate	96.3%	77.7%	−18.6 pp
$\mathcal{D}$	Confused deputy rate	47.0%	2.0%	−45.0 pp
$\mathcal{C}$	Contact enforcement	100%	100%	0
$\mathcal{X}$	Cross-cluster leak rate	87.5%	69.6%	−17.9 pp
$\mathcal{A}$	Amplification factor	1.61	1.55	−0.06

5.3.1 Key Findings

D0 is alarming. Without policy: cross_surface_plant 0.0% defence, agents refuse only 1 in 6 queries they should refuse, 96.3% transitive chains leak, 47% of false delegation claims succeed, 70% of unauthorised write requests execute.

Base policy closes direct attacks (should_refuse +57.2pp, confused_deputy −45pp, cross_surface_plant +94pp) **but not network-emergent attacks** (transitive_risk: still 77.7% leak; cross_cluster: still 69.6% leak). The policy addresses bilateral threats but cannot express “protect your contacts’ data from being relayed to third parties.”

Network amplification is real. $\mathcal{A} > 1.5$ under both conditions, roughly constant—a property of topology, not defence.

Contact enforcement is infrastructure. Non-contact probes achieve 100% under both D0 and D1 via ACL check before the agent is invoked.

Phase 2 dig-further resistance. All four runs achieved 100% resistance on scored Phase 2 probes. Aggregated broad probes trigger built-in safety training even without explicit policy, suggesting single-shot

precision attacks are more dangerous than persistent broad probes.

5.3.2 What Transfers from PACT-PAIR and What Is New

Transfers: Policy dramatically improves security at modest utility cost (4.6:1 ratio). Over-refusal is the primary utility cost. Contact enforcement provides a hard structural floor.

New to PACT-NET: Transitive leakage is a distinct failure mode ($\mathcal{J} = 77.7\%$ even with policy). Cross-surface planting is the most dramatic improvement (+94.0pp). Network amplification ($\mathcal{A} > 1.5$) means dyadic evaluation systematically underestimates real-world leakage. These findings motivate architectural solutions.

Chapter 6

Architectural Solutions

The preceding chapters established what goes wrong. This chapter proposes architectural solutions that address failures not by asking the model to try harder but by restructuring the execution environment so that violations become structurally impossible.

The governing philosophy: *do not ask the agent to refrain from disclosing data; ensure the data is not in the room.*

Table 6.1: Failure modes and their architectural remedies.

Failure mode	Root cause	Remedy
Direct information leak	Data present in context	Mountable Context Cells
Transitive leakage	Co-located sensitive data	Mountable Context Cells
Confused deputy	No delegation verification	Escalation Protocol
Over-refusal	Binary access model	Per-requester MCC

6.1 Relationship-Specific Policy (D3)

Information sensitivity is relational: the same fact may be appropriate for one requester and harmful for another. D3 parameterises governance by requester identity, assembling a tailored system prompt per relationship type. Results (Table 6.2) show D3 achieves 15.5% aggregate leak with 70.9% utility—substantially outperforming flat D2 for misaligned requesters. But within-scope leakage persists (38.7% for Close Friend), motivating the structural solutions below.

6.2 Mountable Context Cells

6.2.1 Concept

A Mountable Context Cell (MCC) is a capability-based execution sandbox assembled per-requester. The analogy is containerisation versus file permissions: containers construct a filesystem view in which unauthorised files *do not exist*. The agent cannot access what is not mounted.

MCC Specification

```

allowedNoteIds: [list of accessible document IDs]

calendarScope: "full" | "busy_only" | "none"

todoFilter: {projects: [...], priorities: [...]}

allowedTools: [list of invocable tool names]

memoryShards: [list of accessible memory shard IDs]

```

Different requesters receive different MCCs. Engineer A sees sprint notes and tech docs with full calendar access. Investor B sees financial summaries with busy-only calendar. The MCC eliminates transitive leakage (co-located data not mounted), over-refusal (agent freely uses everything in context), and cross-cluster conflation (domain-specific access regardless of closeness).

6.2.2 Implementation

MCCs use namespace-based permissions (folder-level access control). Assembly pipeline: (1) relationship identification, (2) policy resolution to MCC template, (3) concrete MCC construction, (4) context injection. Access is granted at folder level; new documents inherit the folder's access profile.

6.3 MCC Experiments & Results

Three-condition factorial design using PACT-PAIR tasks (GPT-5.5, 5 requesters, 400 QA tasks each):

- **D3 (prompt only)**: Full data loaded, per-requester policy in system prompt.
- **MCC_H (structure only)**: Folder-scoped access, no policy prompt.
- **MCC_H_D3 (both)**: Folder-scoped access + per-requester policy.

Table 6.2: Three-condition comparison (6,000 judged responses total).

Requester	Role	D3 (prompt)		MCC_H (structure)		MCC_H+D3 (both)	
		Util	Leak	Util	Leak	Util	Leak
R0	Stranger	21.0	1.2	23.8	1.6	23.9	0.6
R1	Colleague	87.4	8.0	82.9	20.3	83.3	6.3
R2	Delegate	87.1	12.6	86.6	17.9	85.9	12.6
R3	Close Friend	63.3	38.7	26.2	22.0	27.1	18.1
R4	Investor	87.4	16.9	59.7	1.6	60.6	2.7
Aggregate		70.9	15.5	57.6	12.4	58.5	8.0

Four findings:

1. Structure alone reduces leaks for misaligned requesters (R3: -16.7pp , R4: -15.3pp) but *increases* them for aligned ones (R1: $+12.3\text{pp}$) because the agent freely discloses everything within mounted folders.
2. The combined MCC_H+D3 dominates for misaligned requesters: R3 $38.7\%\rightarrow 18.1\%$, R4 $16.9\%\rightarrow 2.7\%$.
3. Prompt policy provides within-scope discrimination that structure cannot (R1: $20.3\%\rightarrow 6.3\%$ from adding D3 to MCC_H).
4. Structural and behavioural governance are complementary, not substitutes.

6.4 Intelligent Escalation Protocol**6.4.1 Design**

MCCs address data-absence enforcement but cannot resolve within-scope boundary cases. The Escalation Protocol operates as a *pre-tool gate*: before any read/write tool executes, the gate classifies the operation as Continue or Stop. Stop prevents execution and creates an escalation record for the owner.

Precedent-based learning. Each owner decision becomes a precedent. Future matching queries are auto-resolved at three levels: exact match, requester-class match (90% threshold), and pattern match. As precedents accumulate, the auto-decision rate rises from 47.1% at 10% precedent to 95.2% at 50%.

6.4.2 Results

Validated on both PACT-PAIR and PACT-NET (12 conditions: $2 \text{ models} \times 2 \text{ tracks} \times 3 \text{ precedent fractions}$, 11,906 gate decisions):

Table 6.3: Escalation gate results (GPT-5.5).

Track	Precedent	PStop	Utility	ContactBlock	Auto-decide
NET	10%	95.1	74.6	97.8	47.1
NET	30%	90.7	91.3	97.1	90.9
NET	50%	91.1	93.2	96.0	95.2
PAIR	10%	90.1	70.7	—	78.0
PAIR	30%	84.3	37.3	—	97.5
PAIR	50%	83.4	23.8	—	99.4

On PACT-NET, the gate achieves $>90\%$ PStop while utility rises above 91% at 30–50% precedent. The PAIR-Relationship track exposes a limitation: P-skewed label distributions cause the auto-decider to over-stop

minority L cases (utility drops to 23.8% at 50% precedent). This is fixable by label-balanced precedent clustering.

6.5 Production Deployment

The solutions are deployed in production (Pulse/Aicoo):

Table 6.4: Deployment status of architectural solutions.

Component	Status	Limitation
Folder-scoped MCC	Deployed	Document-level, not field-level
Per-requester tool scoping	Deployed	Manual configuration
Relationship memory shards	Deployed	L2 only
Pre-tool escalation gate	Deployed	Gate-only, not final-answer
Precedent learning	Deployed	Consensus heuristic

MCCs are realised as folder-scoped share links: each link encodes an MCC specification. The retrieval index is rebuilt per-link to contain only permitted documents. This is not a permission check after retrieval but a scoping constraint before retrieval.

Chapter 7

Conclusion & Discussion

SharedOS is not merely a research prototype: the deployed platform (Pulse/Aicoo) hosts thousands of active agents with thousands of real cross-agent communications daily, grounding our findings in production-scale evidence.

7.1 The Frontier Cannot Be Flattened By Prompt Engineering

Under the best prompt configuration, 27–38% of protected information still leaks in dyadic evaluation. At network scale, transitive leaks persist at 77.7%. Three forces prevent prompt engineering from eliminating the residual:

Implicit social norms. Models resolve ambiguous cases by defaulting to social convention rather than strict policy compliance. No prompt specificity can override a training signal pervading the entire pretraining corpus.

Semantic properties of natural language. A response revealing no single protected fact may collectively reveal protected information through implication or elimination.

Conversational dynamics. Multi-turn interaction creates pressure toward disclosure. PACT-PAIR’s 3× gap between single-turn and multi-turn leakage quantifies this.

Architectural solutions (MCC, Escalation) break through by operating at a different layer: removing data from context or gating tool execution before protected data is retrieved.

7.2 Implications for Agent System Design

Safety training is not privacy governance. Safety addresses universal harms (toxic content is harmful regardless of context). Privacy governance addresses contextual harms (the same content is harmful for one requester and appropriate for another).

Relationship context is a feature, not a bug. Systems treating privacy as a property of data objects will fail. Governance must be parameterised by requester identity (MCCs embody this).

Over-refusal is the dominant real-world failure. Over-refusal accounts for 18–25% of utility failures under D2. Users abandon agents that refuse too often. Security and utility must be optimised jointly.

Read trust $\neq$ write trust. PACT-NET’s cross-surface plant results (+94pp defence with policy) reveal that a single “trust level” granting both read and write access will systematically fail.

7.3 Limitations

Synthetic data. Cannot capture the full complexity of real personal data.

Same-model attacker/defender. May underestimate adversarial capability.

Organic multi-turn only. Does not employ formal adversarial techniques (PAIR, Crescendo, TAP).

Partial MCC validation. Folder-level, not field-level. Todos use coarse on/off gating.

Single ecosystem. Technology startup context; cannot claim generalisation to healthcare or government.

7.4 Summary of Contributions

1. **SharedOS:** First platform for systematic study of cross-boundary agent privacy with persistent state, tool-mediated interactions, and configurable defence mechanisms.
2. **PACT-PAIR:** 600-task dyadic benchmark revealing the specificity threshold (RQ1), bounded multi-turn erosion (RQ2), and relationship-dependent frontiers (RQ3).
3. **PACT-NET:** 997-task network benchmark revealing transitive leakage ($\mathcal{J} = 77.7\%$), network amplification ($\mathcal{A} > 1.5$), and the efficacy of structural enforcement.
4. **Mountable Context Cells:** Structural governance reducing aggregate leaks from 15.5% to 8.0% (combined with prompt policy).
5. **Intelligent Escalation Protocol:** Pre-tool gate achieving >90% PStop with auto-decision rates reaching 95.2%.

7.5 Future Work

Formal attack integration. PAIR, Crescendo, PAP as automated red-teamers within PACT.

Per-field MCC. Sub-document granularity (e.g., calendar event time without title).

Formal verification. Model checking for non-interference guarantees on MCC isolation.

User study. Real participants, genuine relationships, authentic personal information.

Network-scale MCC. Distributed policy propagation maintaining provenance transitively.

7.6 Closing Statement

The central thesis: *do not ask the delegate to refrain from accessing sensitive data; ensure the sensitive data is not in the room.* This transforms privacy from a behavioural property (dependent on LLM compliance) into a structural property (enforced by the execution environment). The coordination problem facing AI agents will not be solved by more capable individual agents—a more capable agent is also a more capable adversary. The solution lies in trust infrastructure between agents: mechanisms enabling safe coordination without requiring perfect alignment, perfect robustness, or perfect obedience. SharedOS is a step toward that infrastructure.

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